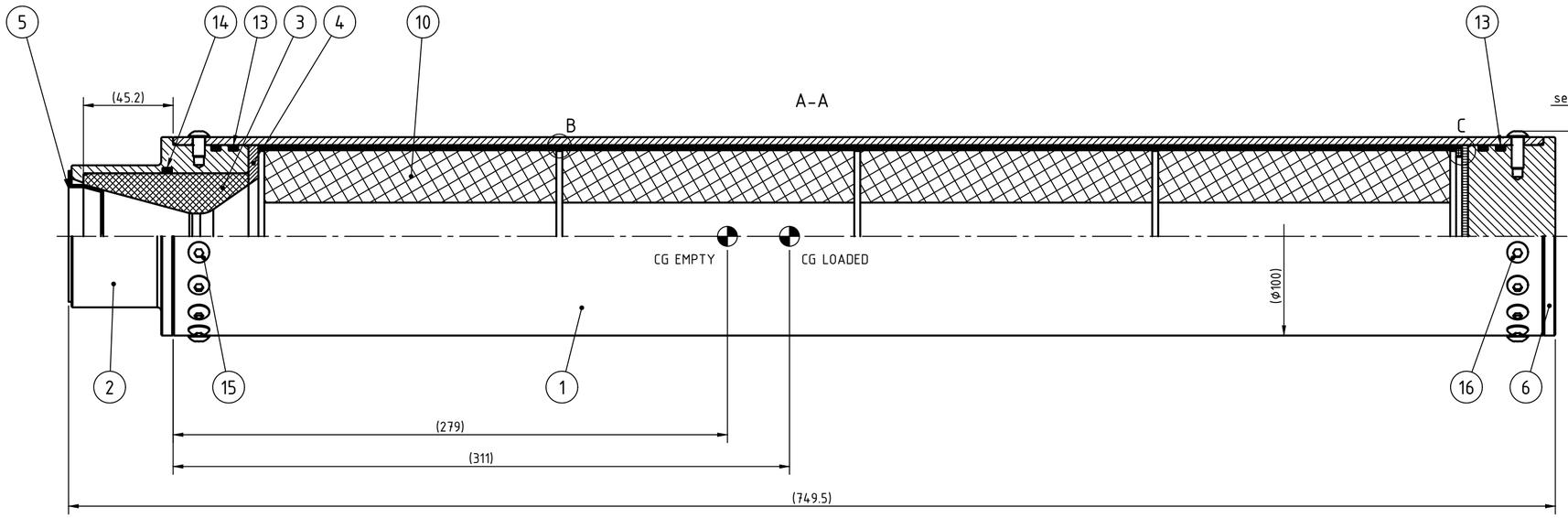
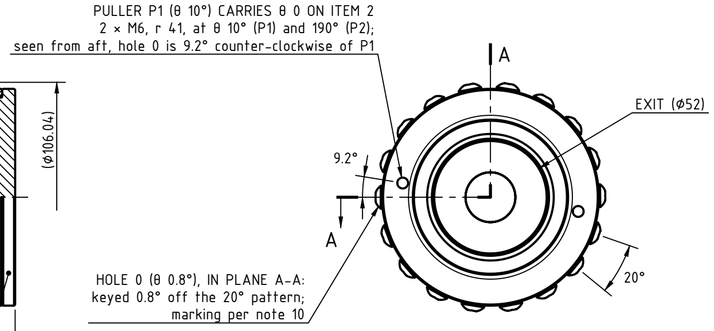
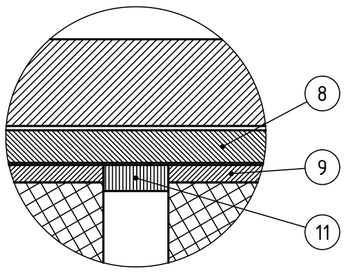




AFT END VIEW (item 5 not shown)



B (5:1)

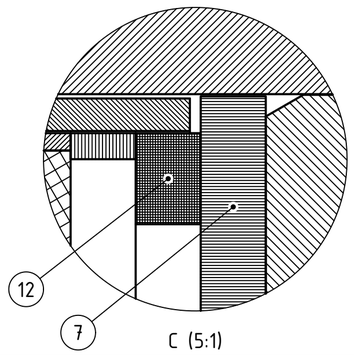
SECTION HATCHING (ISO 128-50)

- Metal, alternating: 1 (45°); 2, 6 (135°)
- Graphite: 3
- Glass phenolic: 4
- Paper phenolic: 8 (detail B)
- Spiral-wound kraft: 9 (detail B)
- Linen phenolic: 11 (detail B)
- Propellant, APCP: 10
- EPDM: 7
- Silicone sponge: 12 (detail C)
- Solid: 13, 14 (O-rings), 5; at 1:2 also 8, 9, 11
- Not sectioned: screws 15, 16

MOTOR DATA

PARAMETER	VALUE (nominal run unless stated)
Propellant	APCP 70/16/14 (AP/Al/HTPB)
Density, as cast	1 718 kg/m³ (98 % of 1 753 theoretical)
Grains	4 × BATES, ø86.8 / ø34 × 147.2 both ends burning
Designation	N2203
Total impulse / class	10 495 N·s / N (2.5 % into N) M/N borderline until a static fire
Thrust, average / peak	2203 / 2383 N
Burn time	4.76 s
Specific impulse	211.2 s
Pc, average / peak	3.77 / 4.05 MPa
Kn, initial / peak	247.8 / 281.1, nominal simulated run (erosive) 279.5 max geometric (uniform regression)
MEOP	7.0 MPa
Hydrostatic proof	10.5 MPa (1.5 × MEOP), SRM100-001
Throat / exit	ø23.00 / ø52.0
Expansion ratio	5.11
Mass, loaded / empty	10.601 / 5.534 kg
Propellant mass	5.067 kg (fraction 0.478)
CG, loaded / empty	311 / 279 mm fwd of the case aft face
Bound run (sizing)	N2523: 11 504 N·s, Pc peak 4.59 MPa (the paper's ideal inputs)

– Nominal run: 1 718 kg/m³ as cast, c* × 0.95, Mukunda-Paul erosive burning, grain at 20 °C. Predictions from srm_ballistics, not measured: the static fire settles the class.



C (5:1)

NOTES

1. Assemble per sheet 2 (aft end: items 2–5, 13–15) and sheet 3 (liner, grain stack and forward closure).
2. Hydrostatic proof per SRM100-001 before the first firing and after any repair, any retired-and-replaced part, or a temperature label reading ≥ 232 °C: 3.5 / 7 / 10.5 MPa, 300 s holds, relief valve 11 MPa, decay ≤ 2 %, permanent set ≤ 0.02 mm. The proof build uses proof dummy SRM100-120 in place of item 3 (item 14 stays); items 4, 5 and 7–12 are not fitted. Never wet or hydro-test the graphite.
3. ENERGETIC MATERIAL: item 10 (APCP) makes this an N-class research motor. Make, store, transport and fire it only as local law allows and under the Tripoli Research (Level 3) and NFPA 1127 rules; minimum distance for N: 1000 ft (≈ 305 m). The site RSO's rules govern.
4. Nozzle cap item 5: fitted at assembly as the debris seal, taped per SRM100-106 note 6, kept on through storage and transport. At the stand it comes off only to insert the igniter (leads shunted, out of its ø3 hole), then it is refitted and re-taped with new tape. It blows out at ignition.
5. The igniter is not part of this drawing (README: 3 g BKNO3 or pyrogen, tube ≤ ø10 × 40, dual-bridge e-match). It goes in last, at the stand, leads shunted. Never move the motor with an igniter in it.
6. Screws, new ISO 7380-1 010.9 every firing: 18 × item 15 at 3–3.5 N·m, nickel anti-seize on the threads only, head face dry; 18 × item 16 at 4–6 N·m with Loctite 243. Tighten criss-cross (opposite pairs) in 3 passes, sheets 2 and 3. Never interchange the lengths: an M6×16 in item 2 bottoms with its head proud, an M6×10 in item 6 engages only 1 d.
7. O-rings 13, 14: FKM 90A (Parker V0709-90 or equal), new every firing. Lube them, the liner OD and the insulator edges with Parker Super O-Lube or DC-55, thin film. Bond item 7 to item 6 with RTV106 or equal on SS4004P primer, bondline 0.1–0.2 (sheet 3).
8. Seal zones of item 1, both ends: each honed to the smallest diameter that cleans it up, A (the smallest reading in the zone, recorded per zone, SRM100-101 note 4), 92.000 ≤ A ≤ 92.070, held to A +0.030/0, Ra 0.4 (0.8 max); the bore is ≤ 92.10 everywhere. Items 2 and 6 are cut to their own case zone's A: land A – 0.06 0/–0.02, groove bottom A – 5.67 0/–0.04.
9. Each case end and its closure are a match-drilled, serialised pair. θ 0 is carried on item 1 by hole 0, the scribed clocking line and the AFT / FWD engraving; on items 2 and 6 by puller P1 at θ 10° (r 41 on item 2, r 37.5 on item 6). Hole 0 (θ 0.8°) is 9.2° clockwise of P1 seen from forward and 9.2° counter-clockwise seen from aft (end view); keyed 0.8° off the 20° pattern, a pair goes together one way only.
10. Marking and temperature labels per SRM100-101 note 8 (item 1) and SRM100-102 note 8 (item 2); marking of item 6 per SRM100-103 note 11. Labels: irreversible 177 / 204 / 232 / 260 °C strips, new for every firing, at θ 90° and 270° on the item 1 OD at z 20–35 and the item 2 aft body ø71.58, clear of the cap tape. After firing read and record them before extraction; ≥ 232 °C: retire or re-proof that end (note 2).
11. No pressure port: item 6 is solid (no port, no pocket) and item 7 is a full disc. The motor is fired without a chamber transducer; the static-fire record is the load cell. A transducer needs a new, ported item 6 and a re-proof. The proof fills and bleeds through SRM100-120's port (SRM100-001).
12. Liner item 8: cut to the measured cavity minus 0.5 float. Check the float by depth gauging before item 6 goes in, not with a feeler (the gap is closed off). Take-up ring item 12: 3.2, 4.8 or 6.4 sheet chosen from the measured gap for 25–50 % compression. Both per sheet 3.

NOTES (continued)

13. After every firing replace items 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, all 36 screws (items 15, 16) and the temperature labels; re-use items 1, 2, 6 after inspection (SRM100-101 R10 for the case), and item 3 while the throat is ≤ ø23.9 and uncracked.
14. Static fire: item 6's forward face is the thrust face, flat from the axis to ø100, broken only by the 2 M6 puller holes at r 37.5; a load cell or button bears anywhere clear of them. Flight: the airframe needs a ø108 keep-out for the screw heads and a thrust ring or a forward-closure thrust path.
15. All dimensions on this sheet are reference dimensions. Masses and CG are from the STEP solids (mass_budget.json); CG measured forward from the case's aft end face.
16. Tooling, referenced by name only: acetal hole plugs, push sleeve, casting mandrel, puller bridge with a split clamp collar (100.0 bore) on the case OD ≥ 50 from the end being extracted, clear of the screw heads 13 from each end face (ø106.04 sweep), the labels and the marking 20–45 from each end face.
17. Proof dummy SRM100-120 and proof assembly SRM100-001 are proof hardware only: never fired, not part of this assembly.

USED ON: HELIOS rocket – SRM-100 motor (static-fire stand, then flight)

ITEM	QTY	PART NO. / STANDARD	DESCRIPTION	MATERIAL	MASS g (each)
–	4	Temp. label 177–260 °C	On 1, 2: new each firing (note 10)	irreversible strip	–
–	a/r	PVC tape, 19 wide	Cap tape: 5 (SRM100-106 note 6)	PVC electrical tape	–
–	a/r	RTV106 + SS4004P	Bond 7 to 6, primed (note 7)	RTV silicone + primer	–
–	a/r	Loctite 243; anti-seize	Threads: 16 (243), 15 (anti-seize)	anaerobic / nickel	–
–	a/r	Super O-Lube or DC-55	Lube: 13, 14, liner OD (note 7)	silicone, thin film	–
16	18	ISO 7380-1 M6×16-010.9	Button-head screw, fwd	steel 010.9	4.7
15	18	ISO 7380-1 M6×10-010.9	Button-head screw, aft	steel 010.9	3.4
14	1	AS568-230	O-ring, insert seal	FKM 90A	3.8
13	4	AS568-237	O-ring, case seal	FKM 90A	5.1
12	1	SRM100-108	Take-up ring	silicone sponge	1.7
11	5	SRM100-109	Grain spacer	linen phenolic	1.3
10	4	SRM100-112	Propellant grain, BATES	APCP	1266.8
9	4	SRM100-111	Casting tube	spiral-wound kraft	30.8
8	1	SRM100-110	Liner	paper phenolic	347.5
7	1	SRM100-107	Forward insulator	EPDM	22.8
6	1	SRM100-103	Forward closure	6061-T651	770.5
5	1	SRM100-106	Nozzle cap	LDPE	6.6
4	1	SRM100-105	Nozzle insulator	glass phenolic	38.8
3	1	SRM100-104	Nozzle insert	isomolded graphite	311.1
2	1	SRM100-102	Nozzle holder (aft closure)	304 stainless	1501.2
1	1	SRM100-101	Motor case	6061-T6 drawn tube	2234.4

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
See parts list	10.60 kg	ISO 2768-mK	–	–

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
ASSEMBLY DRAWING	NOT CHECKED – FOR REVIEW	1:2		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	GENERAL ARRANGEMENT SRM-100 motor, loaded – parts list, motor data

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-000	A	1 / 3